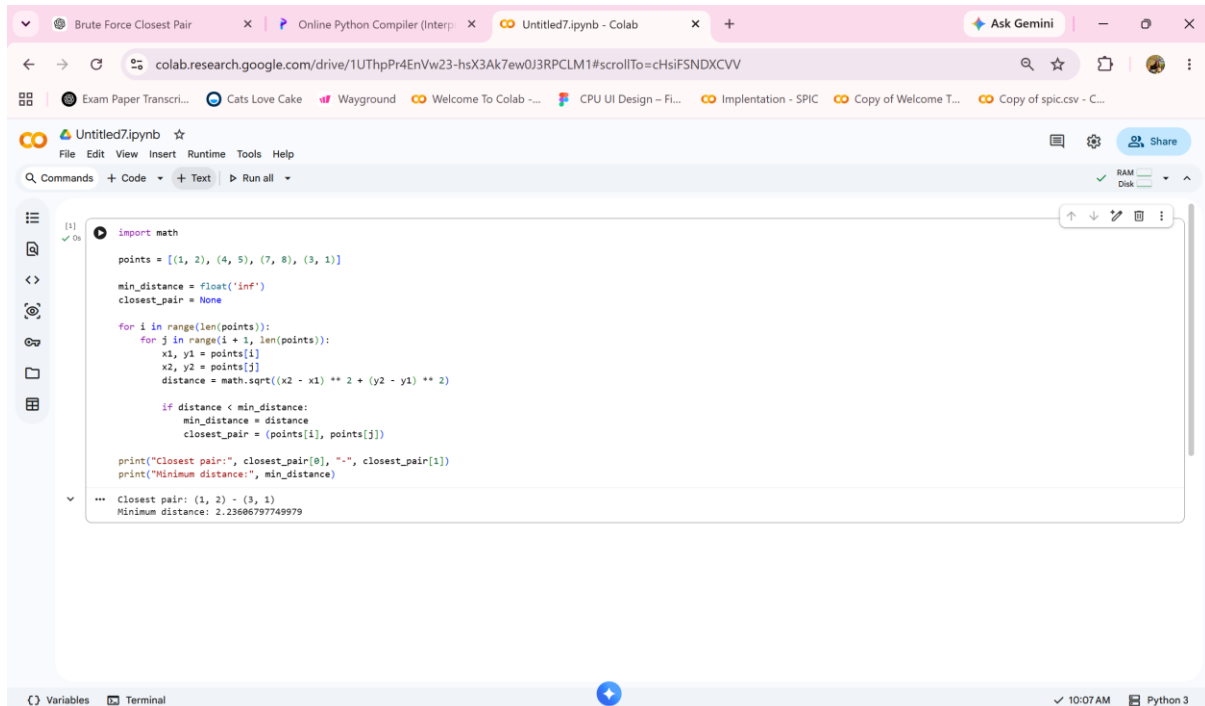


LAB UNIT-4



Brute Force Closest Pair

Online Python Compiler (Interp...

Untitled7.ipynb - Colab

Ask Gemini

colab.research.google.com/drive/1UThpPr4EnVw23-hsX3Ak7ew0J3RPCLM1#scrollTo=cHsiFSNDXCvV

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Untitled7.ipynb

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```
[1] ✓ 0s
import math

points = [(1, 2), (4, 5), (7, 8), (3, 1)]

min_distance = float('inf')
closest_pair = None

for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]
        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)

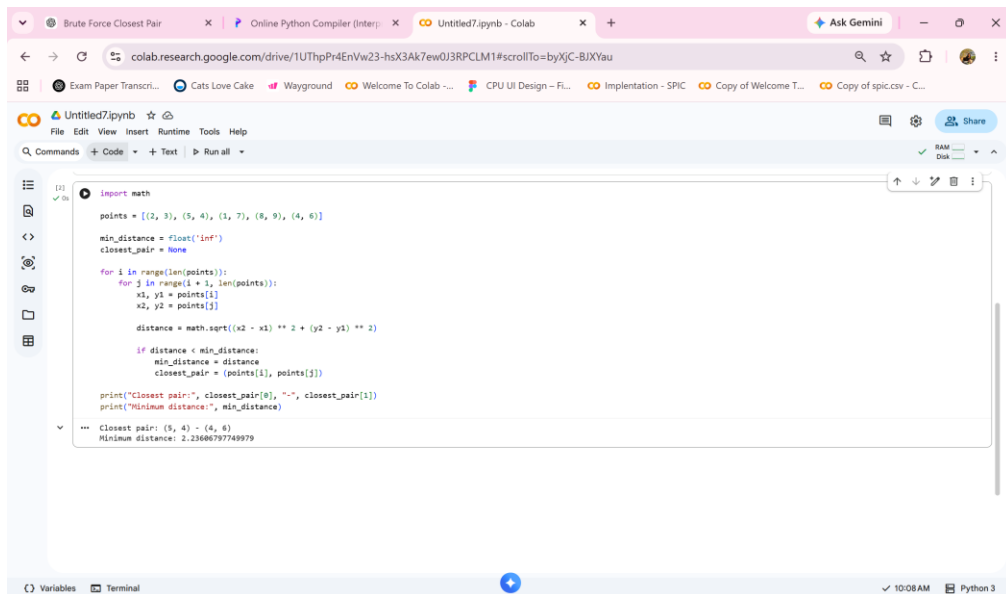
        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])

print("Closest pair:", closest_pair[0], "-", closest_pair[1])
print("Minimum distance:", min_distance)
```

Closest pair: (1, 2) - (3, 1)
Minimum distance: 2.23606797749979

Variables Terminal

10:07 AM Python 3



Brute Force Closest Pair

Online Python Compiler (Interp...

Untitled7.ipynb - Colab

Ask Gemini

colab.research.google.com/drive/1UThpPr4EnVw23-hsX3Ak7ew0J3RPCLM1#scrollTo=byXjC-BjXVau

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Untitled7.ipynb

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```
[1] ✓ 0s
import math

points = [(2, 3), (5, 4), (1, 7), (8, 9), (4, 6)]

min_distance = float('inf')
closest_pair = None

for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]

        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)

        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])

print("Closest pair:", closest_pair[0], "-", closest_pair[1])
print("Minimum distance:", min_distance)
```

Closest pair: (5, 4) - (4, 6)
Minimum distance: 2.23606797749979

Variables Terminal

10:08 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab x + Ask Gemini

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Untitled7.ipynb

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Commands + Code + Text Run all

```
[1] ✓ Os
min_distance = float('inf')
closest_pair = None

for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]

        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)

        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])

print("Closest pair:", closest_pair[0], "-", closest_pair[1])
print("Minimum distance:", min_distance)
```

Closest pair: (2, 2) - (3, 1)
Minimum distance: 1.4142135623730951

Variables Terminal 10:09 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab x + Ask Gemini

colab.research.google.com/drive/1UThpPr4EnVw23-hsX3Ak7ew0J3RPCLM1#scrollTo=ebZs8kXFXwBT

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```
[3] ✓ Os
print("Closest pair:", closest_pair[0], "-", closest_pair[1])
print("Minimum distance:", min_distance)
```

Closest pair: (2, 2) - (3, 1)
Minimum distance: 1.4142135623730951

```
[4] ✓ Os
import math

points = [(10, 20), (15, 25), (30, 40), (12, 22)]

min_distance = float('inf')
closest_pair = None

for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]

        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)

        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])

print("Closest pair:", closest_pair[0], "-", closest_pair[1])
print("Minimum distance:", min_distance)
```

Closest pair: (30, 20) - (12, 22)
Minimum distance: 2.8284271247461903

Variables Terminal 10:10 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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Untitled7.ipynb

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Commands + Code + Text Run all

```
def euclidean_distance(p1, p2):
    return math.sqrt((p2[0]-p1[0])**2 + (p2[1]-p1[1])**2)

def closest_pair(points):
    min_distance = float('inf')
    pair = None

    for i in range(len(points)):
        for j in range(i + 1, len(points)):
            distance = euclidean_distance(points[i], points[j])

            if distance < min_distance:
                min_distance = distance
                pair = (points[i], points[j])

    return pair, min_distance

points = [(1,2), (4,5), (7,8), (3,1)]
pair, distance = closest_pair(points)

print("Closest Pair:", pair[0], "-", pair[1])
print("Minimum Distance:", distance)

print("\nTime Complexity: O(n^2)")
print("Space Complexity: O(1)")
```

Closest Pair: (1, 2) - (3, 1)
Minimum Distance: 2.23606797749979
Time Complexity: O(n^2)
Space Complexity: O(1)

Variables Terminal

10:10 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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Untitled7.ipynb

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```
if k == i or k == j:
    continue

val = orientation(points[i], points[j], points[k])

if val > 0:
    left += 1
elif val < 0:
    right += 1

if left == 0 or right == 0:
    if points[i] not in hull:
        hull.append(points[i])
    if points[j] not in hull:
        hull.append(points[j])

print("Convex Hull Points:")
for point in hull:
    print(point)
```

Convex Hull Points:
(10, 0)
(5, 3)
(15, 3)
(6, 6.5)
(12.5, 7)

Variables Terminal

Brute Force Closest Pair x Untitled7.ipynb - Colab + Ask Gemini

colab.research.google.com/drive/1UThpPr4EnVw23-hsX3Ak7ew0J3RPCLM1#scrollTo=eNnXOdz-Y9AH

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Untitled7.ipynb File Edit View Insert Runtime Tools Help

Commands + Code + Text + Run all

```
[7] ✓ 0s
def convex_hull(points):
    points = sorted(points)

    lower = []
    for p in points:
        while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
            lower.pop()
        lower.append(p)

    upper = []
    for p in reversed(points):
        while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
            upper.pop()
        upper.append(p)

    return lower[:-1] + upper[::-1]

points = [(0,0), (1,1), (2,2), (0,2), (2,0)]
hull = convex_hull(points)

print("Convex Hull Points:")
for p in hull:
    print(p)
```

Convex Hull Points:
(0, 0)
(2, 0)
(2, 2)
(0, 2)

Variables Terminal 10:15 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab + Ask Gemini

colab.research.google.com/drive/1UThpPr4EnVw23-hsX3Ak7ew0J3RPCLM1#scrollTo=1IQ3DjRcZHBS

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```
[8] ✓ 0s
def convex_hull(points):
    points = sorted(points)

    lower = []
    for p in points:
        while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
            lower.pop()
        lower.append(p)

    upper = []
    for p in reversed(points):
        while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
            upper.pop()
        upper.append(p)

    return lower[:-1] + upper[::-1]

points = [(1,2), (3,1), (5,3), (4,6), (2,5)]
hull = convex_hull(points)

print("Convex Hull Points:")
for p in hull:
    print(p)
```

Convex Hull Points:
(1, 2)
(3, 1)
(5, 3)
(4, 6)
(2, 5)

Variables Terminal 10:16 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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```
[7] ✓ 0s
def convex_hull(points):
    points = sorted(points)

    lower = []
    for p in points:
        while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
            lower.pop()
        lower.append(p)

    upper = []
    for p in reversed(points):
        while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
            upper.pop()
        upper.append(p)

    return lower[:-1] + upper[::-1]

points = [(0,0), (1,1), (2,2), (0,2), (2,0)]
hull = convex_hull(points)

print("Convex Hull Points:")
for p in hull:
    print(p)
```

Convex Hull Points:
(0, 0)
(2, 0)
(2, 2)
(0, 2)

Variables Terminal 10:15 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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```
[9] ✓ 0s
(4, 6)
(2, 5)

def cross(o, a, b):
    return (a[0]-o[0])*(b[1]-o[1]) - (a[1]-o[1])*(b[0]-o[0])

def convex_hull(points):
    points = sorted(points)

    lower = []
    for p in points:
        while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
            lower.pop()
        lower.append(p)

    upper = []
    for p in reversed(points):
        while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
            upper.pop()
        upper.append(p)

    return lower[:-1] + upper[::-1]

points = [(0,3), (1,1), (2,2), (4,4), (3,0)]
hull = convex_hull(points)

print("Convex Hull Points:")
for p in hull:
    print(p)
```

Convex Hull Points:
(0, 3)
(1, 1)
(3, 0)
(4, 4)

Variables Terminal 10:17 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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```
[11] ✓ 0s
lower = []
for p in points:
    while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
        lower.pop()
    lower.append(p)

upper = []
for p in reversed(points):
    while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
        upper.pop()
    upper.append(p)

return lower[:-1] + upper[::-1]

points = [(2,2), (4,1), (6,3), (5,5), (3,6), (1,4)]
hull = convex_hull(points)

print("Convex Hull Points:")
for p in hull:
    print(p)
```

Convex Hull Points:
(1, 4)
(2, 2)
(4, 1)
(6, 3)
(5, 5)
(3, 6)

Variables Terminal

✓ 10:17 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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```
[15] ✓ 0s
def euclidean_distance(p1, p2):
    return math.sqrt((p2[0]-p1[0])**2 + (p2[1]-p1[1])**2)

def closest_pair(points):
    min_distance = float('inf')
    pair = None

    for i in range(len(points)):
        for j in range(i + 1, len(points)):
            distance = euclidean_distance(points[i], points[j])

            if distance < min_distance:
                min_distance = distance
                pair = (points[i], points[j])

    return pair, min_distance

points = [(1,2), (4,5), (7,8), (3,1)]
pair, distance = closest_pair(points)

print("Closest Pair:", pair[0], "-", pair[1])
print("Minimum Distance:", distance)

print("\nTime Complexity: O(n^2)")
print("Space Complexity: O(1)")
```

Closest Pair: (1, 2) - (3, 1)
Minimum Distance: 2.23606797749979
Time Complexity: O(n^2)
Space Complexity: O(1)

Variables Terminal

✓ 10:10 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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```
[12] ✓ 0s
while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
    lower.pop()
    lower.append(p)

upper = []
for p in reversed(points):
    while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
        upper.pop()
        upper.append(p)

return lower[:-1] + upper[:-1]

points = [(1,1), (2,4), (5,5), (6,2), (3,0)]
hull = convex_hull(points)
print("Convex Hull Points:")
for p in hull:
    print(p)

*** Convex Hull Points:
(1, 1)
(3, 0)
(6, 2)
(5, 5)
(2, 4)
```

Variables Terminal

10:18 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab

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Untitled7.ipynb

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Commands + Code + Text Run all

```
[7] ✓ 0s
def convex_hull(points):
    points = sorted(points)

    lower = []
    for p in points:
        while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
            lower.pop()
            lower.append(p)

    upper = []
    for p in reversed(points):
        while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
            upper.pop()
            upper.append(p)

    return lower[:-1] + upper[:-1]

points = [(0,0), (1,1), (2,2), (0,2), (2,0)]
hull = convex_hull(points)
print("Convex Hull Points:")
for p in hull:
    print(p)

*** Convex Hull Points:
(0, 0)
(2, 0)
(2, 2)
(0, 2)
```

Variables Terminal

10:15 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab x + Ask Gemini

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```
[1] ✓ On
def convex_hull(points):
    points = sorted(points)

    lower = []
    for p in points:
        while len(lower) >= 2 and cross(lower[-2], lower[-1], p) <= 0:
            lower.pop()
        lower.append(p)

    upper = []
    for p in reversed(points):
        while len(upper) >= 2 and cross(upper[-2], upper[-1], p) <= 0:
            upper.pop()
        upper.append(p)

    return lower[:-1] + upper[::-1]

points = [(0,0), (1,1), (2,2), (0,2), (2,0)]

hull = convex_hull(points)

print("Convex Hull Points:")
for p in hull:
    print(p)
```

Convex Hull Points:
(0, 0)
(2, 0)
(2, 2)
(0, 2)

Variables Terminal 10:15 AM Python 3

Brute Force Closest Pair x Untitled7.ipynb - Colab x + Ask Gemini

colab.research.google.com/drive/1UThpPr4EnVw23-hsX3Ak7ew0J3RPCLM1#scrollTo=rF0_q-6bXiw-

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Untitled7.ipynb File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
[1] ✓ On
min_distance = float('inf')
closest_pair = None

for i in range(len(points)):
    for j in range(i + 1, len(points)):
        x1, y1 = points[i]
        x2, y2 = points[j]

        distance = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)

        if distance < min_distance:
            min_distance = distance
            closest_pair = (points[i], points[j])

print("Closest pair:", closest_pair[0], "-", closest_pair[1])
print("Minimum distance:", min_distance)
```

Closest pair: (2, 2) - (3, 1)
Minimum distance: 1.4142135623730951

Variables Terminal 10:09 AM Python 3